Social Networks Analysis

MSIS 521

Professor Hossein Ghasemkhani

FOSTER SCHOOL OF BUSINESS
UNIVERSITY OF WASHINGTON

What Is Social Network?

- A map of relationships (formal or informal) among actors (persons, organizations, etc.)
- Representations
  - Graph
  - Matrix (Sociomatrix)

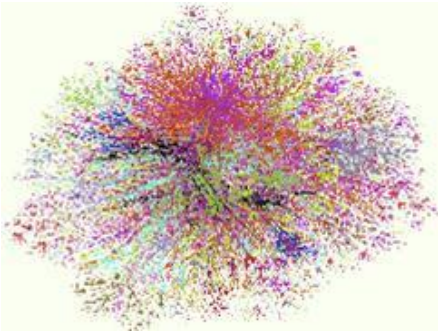
FOSTER SCHOOL OF BUSINESS

Social Network Analysis (SNA)

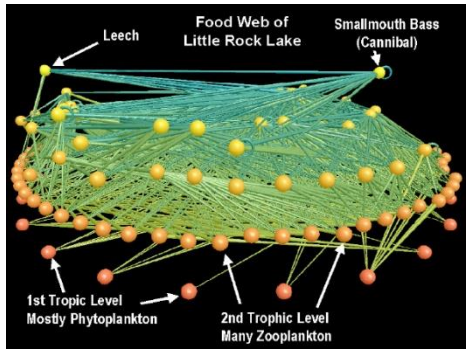
- Social Network Analysis views relationships (knowledge sharing, etc.) in terms of network theory consisting of nodes and ties
- Individuals/actors are represented by nodes
- Relationships are represented by ties/edges
- Ties/edges can be directional
- Several tools are available to draw social network using interaction data: UCINET, PAJEK, R
  - Python Networkx

FOSTER SCHOOL OF BUSINESS

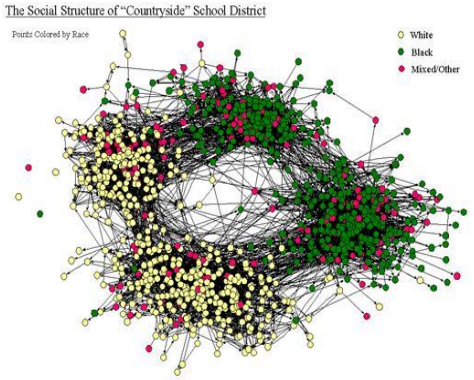
Network Examples



Internet Map
[lumeta.com]

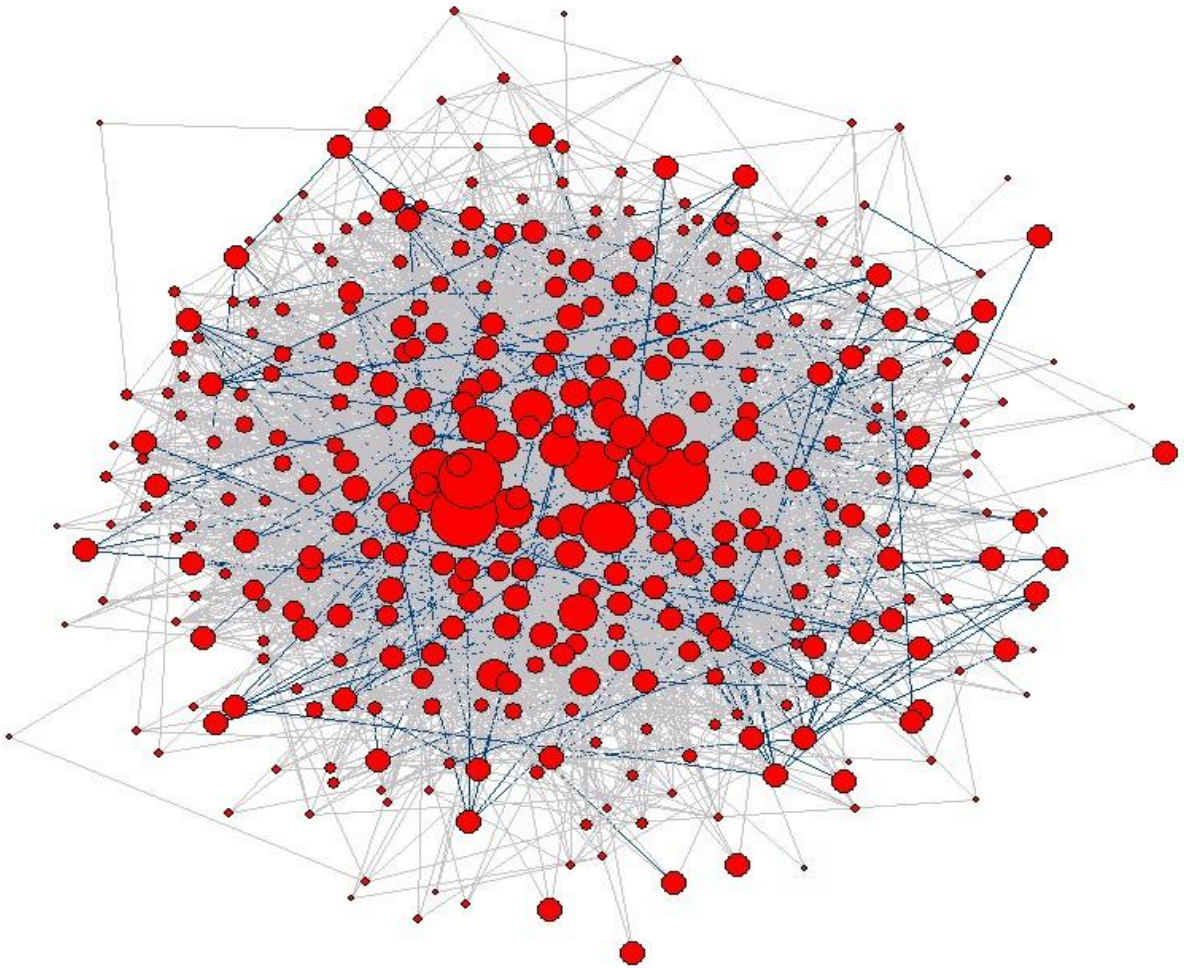


Food Web
[Martinez 91]



Friendship Network
[Moody 01]

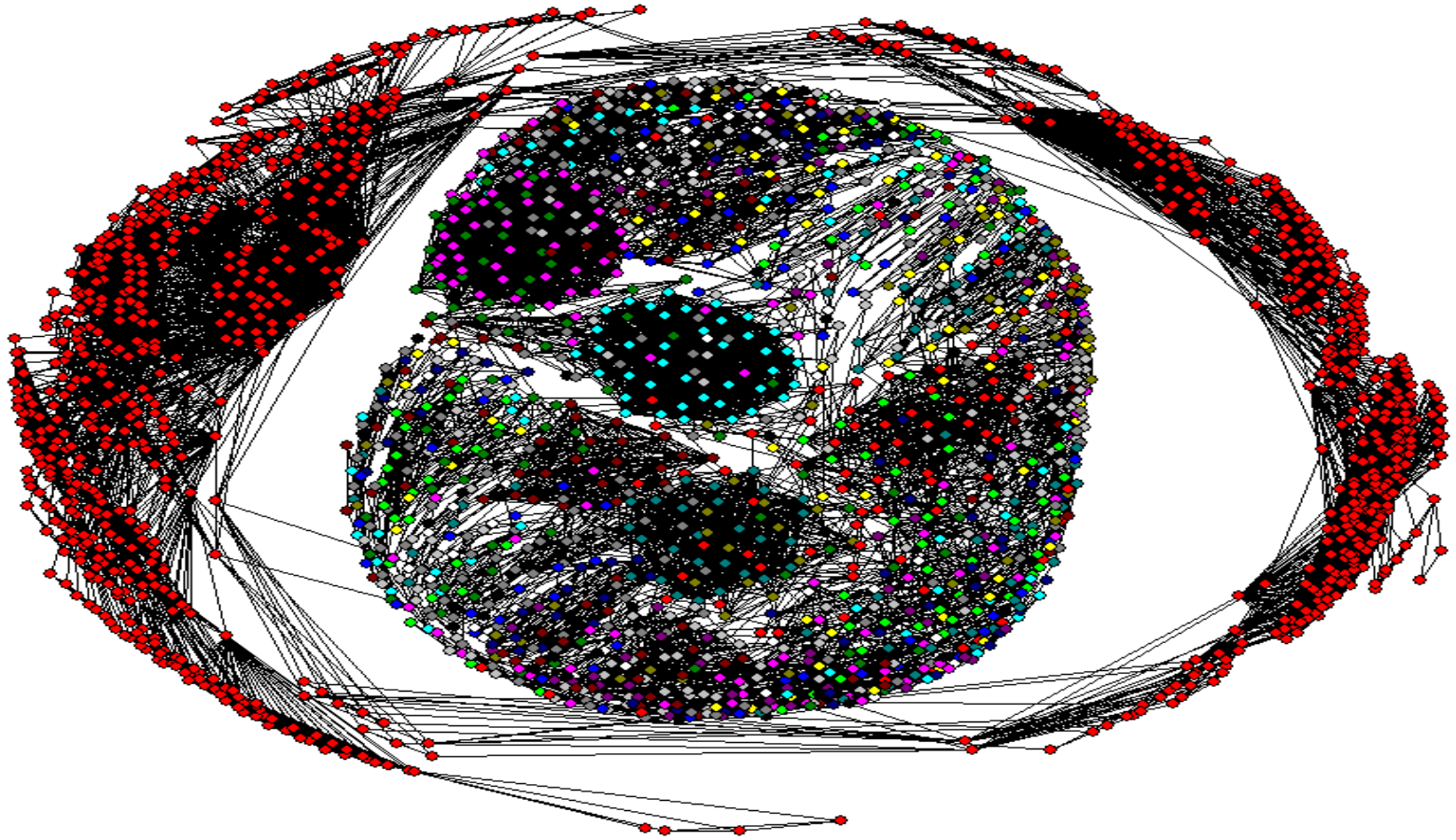
An Example of Friendship Network on Social Media



- Oh, Susarla, and Tan 2012

FOSTER SCHOOL OF BUSINESS

Open-Source Software Collaboration Network



- Singh, Tan, and Mookerjee 2007

FOSTER SCHOOL OF BUSINESS

Why Should We Care?

- Economic actions are “embedded in concrete, ongoing systems of social relations” Granovetter (1985)

- Social Relationships facilitate
  - resource sharing
  - information dissemination
  - and knowledge spillover

FOSTER SCHOOL OF BUSINESS

Where Do We See Networks in Tech & Marketing?

- Many digital systems can be represented as networks (graphs):
  - Social media graphs: people connected by follows, friends, mentions, replies
  - Follower/fan networks around influencers or brands
  - User-item graphs: customers connected to products they view, click, or purchase
  - Interaction graphs: retweets, reposts, comments, likes, shares
  - Knowledge graphs: entities (people, places, products) linked by relationships

- These graphs are the structures that modern graph ML and GNNs operate on.

FOSTER SCHOOL OF BUSINESS

Node-Level Importance: Network Centrality

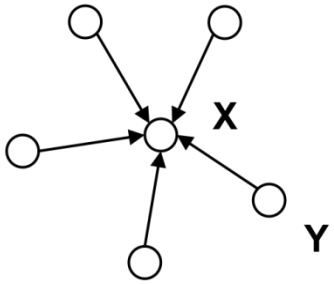
- Which nodes are most ‘central’?
- Definition of ‘central’ varies by context/purpose
- Local measure:
  - Degree
- Relative to the rest of network:
  - Closeness, eccentricity, betweenness, eigenvector (Bonacich power centrality), Katz, PageRank, …
- How evenly is centrality distributed among nodes?
  - Centralization, hubs and authorities, …

FOSTER SCHOOL OF BUSINESS

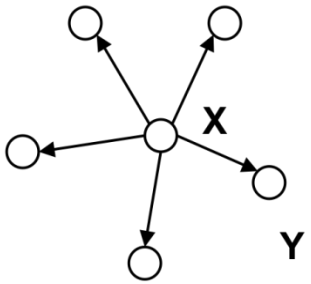
Centrality

Who's important based on their network position

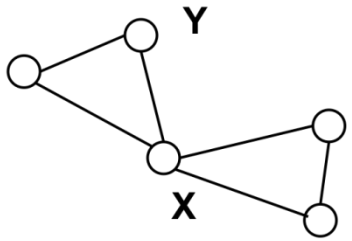
In each of the following networks, X has higher centrality than Y according to a particular measure



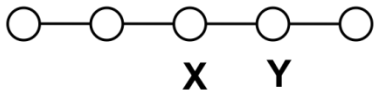
indegree



outdegree



betweenness



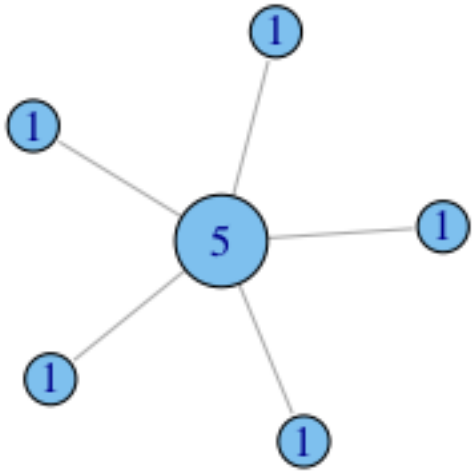
closeness, eccentricity

There are other measures such as eigenvector/PageRank, Katz, etc.

FOSTER SCHOOL OF BUSINESS

Degree Centrality (Undirected)

He who has many friends is most important.



When is the number of connections the best centrality measure?

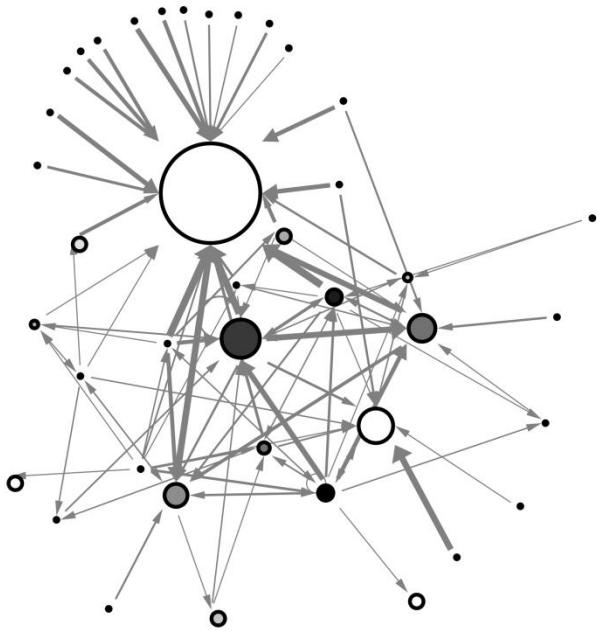
- people who will do favors for you
- people you can talk to (influence set, information access, …)
- influence of an article in terms of citations (using in-degree)

"How many people/items does this node touch?" → used as a **feature** for link prediction, fraud detection (unusually high degree), "influencer" ranking

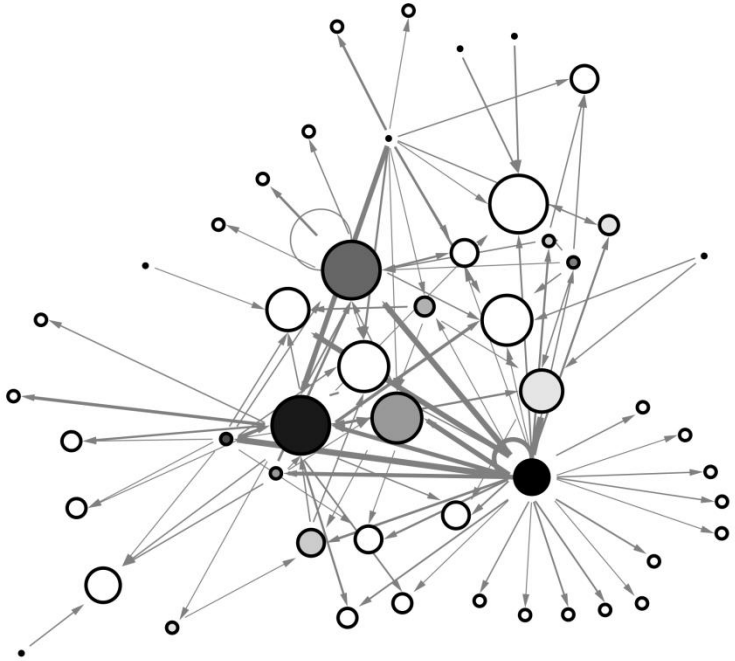
Degree Centralization -- Examples

Example: financial trading networks

Size of node represents degree centrality



high centralization: one node trading with many others

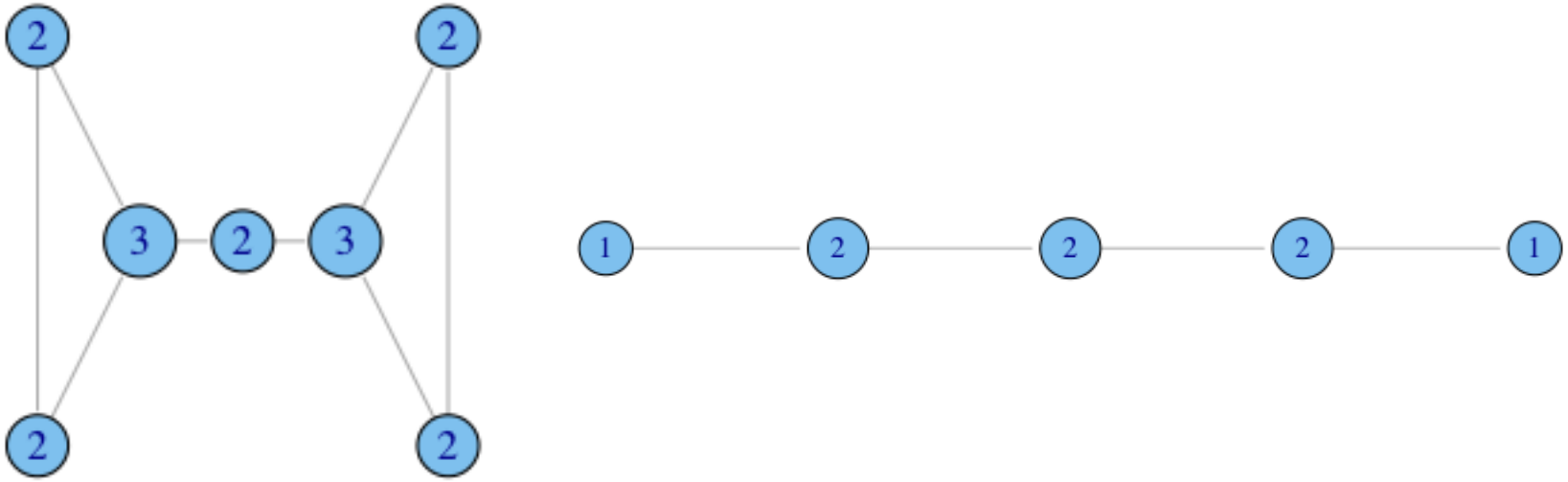


low centralization: trades are more evenly distributed

FOSTER SCHOOL OF BUSINESS

When Degree Isn't Everything

In what ways does degree fail to capture centrality in the following graphs?



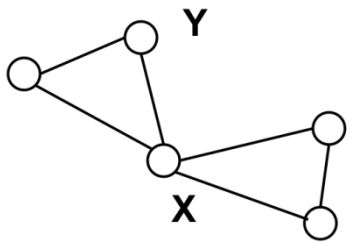
- Ability to broker between groups
- Likelihood that information originating anywhere in the network reaches you…

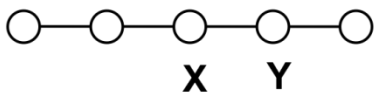
FOSTER SCHOOL OF BUSINESS

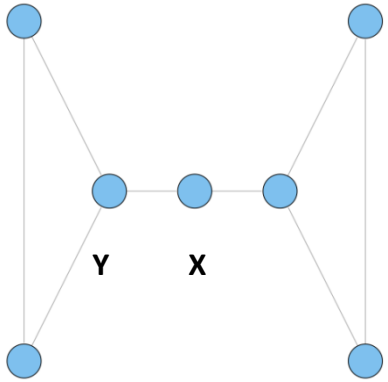
Betweenness: Another Centrality Measure

- Intuition: how many pairs of individuals would have to go through you in order to reach one another in the minimum number of hops?

- Who has higher betweenness, X or Y?



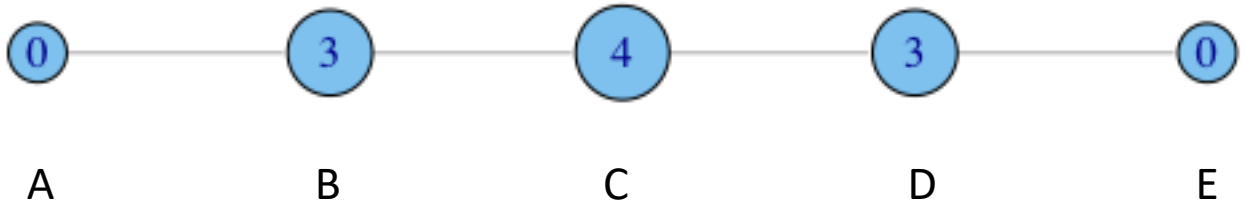




brokerage/bridge nodes → interpretability for **information diffusion models** (who to seed for a campaign)

Betweenness on Toy Networks

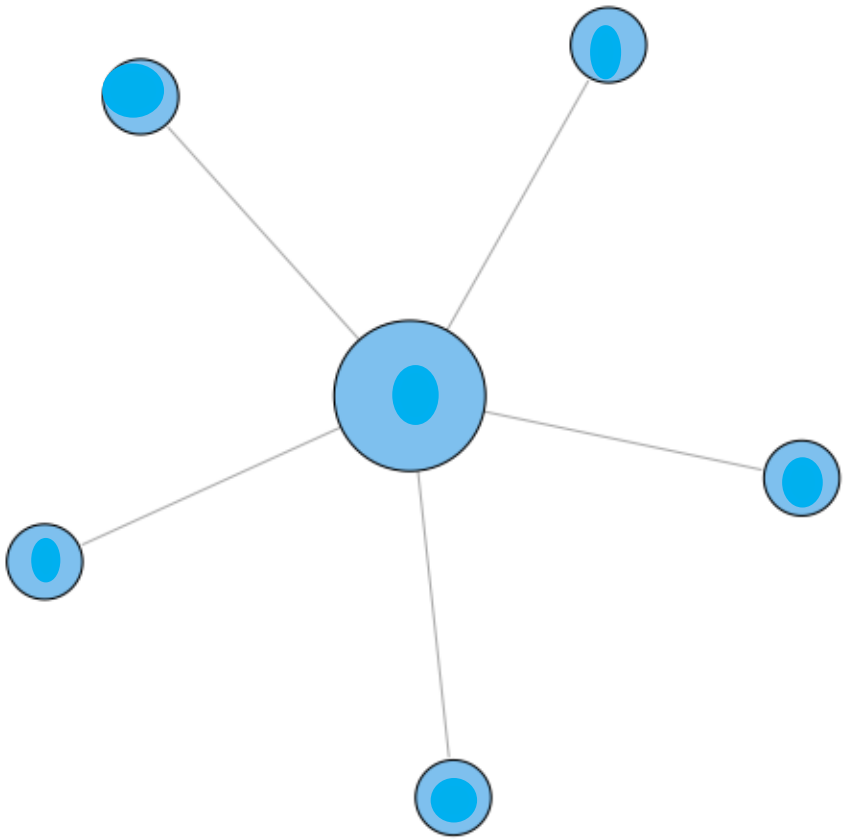
- Non-normalized version:



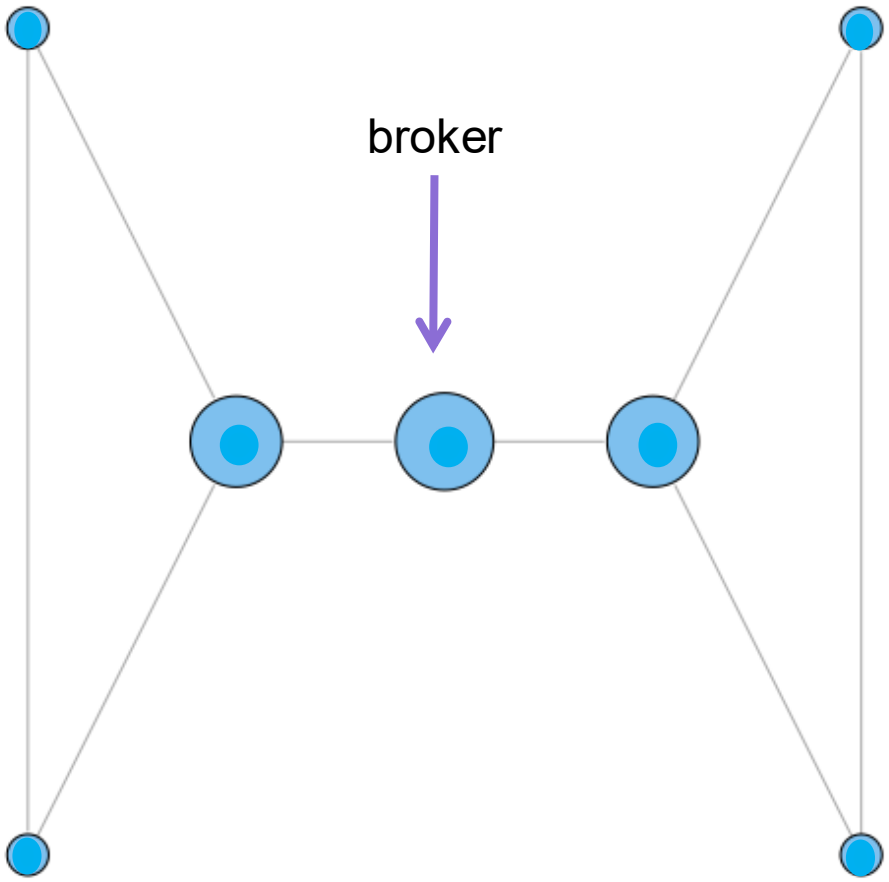
  - A lies between no two other vertices
  - B lies between A and 3 other vertices: C, D, and E
  - C lies between 4 pairs of vertices (A,D),(A,E),(B,D),(B,E)

  - Note that there are no alternate paths for these pairs to take, so C gets full credit.

Betweenness on Toy Networks



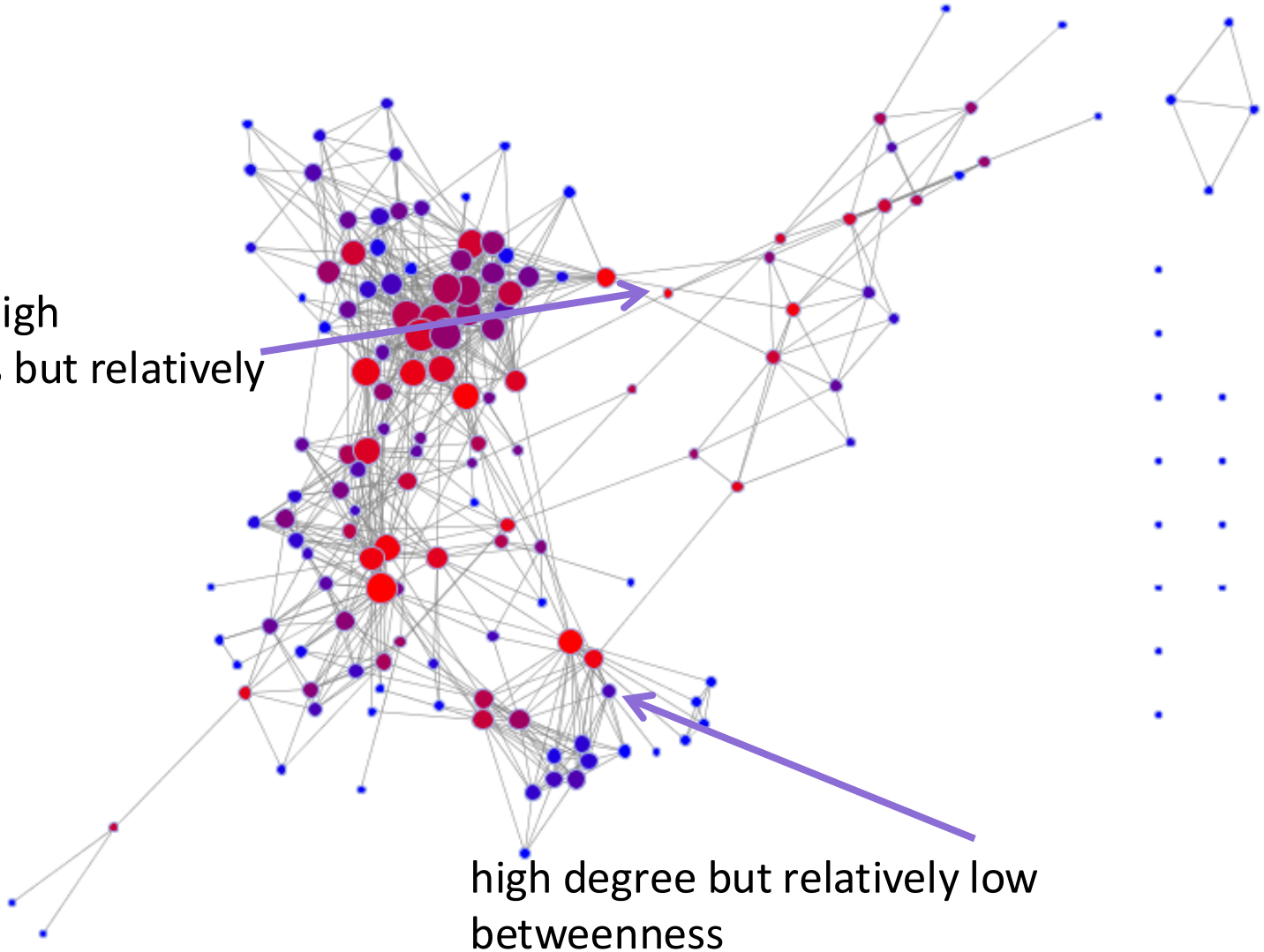
Betweenness on Toy Networks



FOSTER SCHOOL OF BUSINESS

Example

Nodes are sized by degree, and colored by betweenness.



Nodes with high betweenness but relatively low degree

high degree but relatively low betweenness

FOSTER SCHOOL OF BUSINESS

Closeness and Eccentricity

- What if it’s not so important to have many direct friends?
- Or be “between” others.
- But one still wants to be in the “middle” of things, not too far from the center

Closeness is the inverse of the **average** shortest path between a vertex and all other vertices in the graph.

Eccentricity is the **maximum** shortest path between a vertex and all other vertices in the graph.

Closeness measures a node's efficiency in spreading information, while eccentricity identifies nodes with the shortest "worst-case" distance to others. They are used for ranking, optimizing communication, and strengthening network resilience.

FOSTER SCHOOL OF BUSINESS

Exercise: Who Is Most Central?

- Consider this small undirected network with 7 nodes: **A, B, C, D, E, F, G**
- **Edges (connections):**
  - A __ B, A __ C
  - B __ D, C __ D
  - D __ E, D __ F
  - E__G
- **Step 1:** sketch this network on paper.
- **Step 2:**
  - Which node has highest **degree centrality**?
  - Which node has highest **betweenness centrality** (acts as a bridge)?

FOSTER SCHOOL OF BUSINESS

Exercise: Who Is Most Central?

FOSTER SCHOOL OF BUSINESS

Network-Level Patterns

- The Small World
- The Scale Free Networks
- The Strength of Weak Ties
- Cohesive Subgroups
  - Cliques
  - Core/Periphery Structures

FOSTER SCHOOL OF BUSINESS

Small World

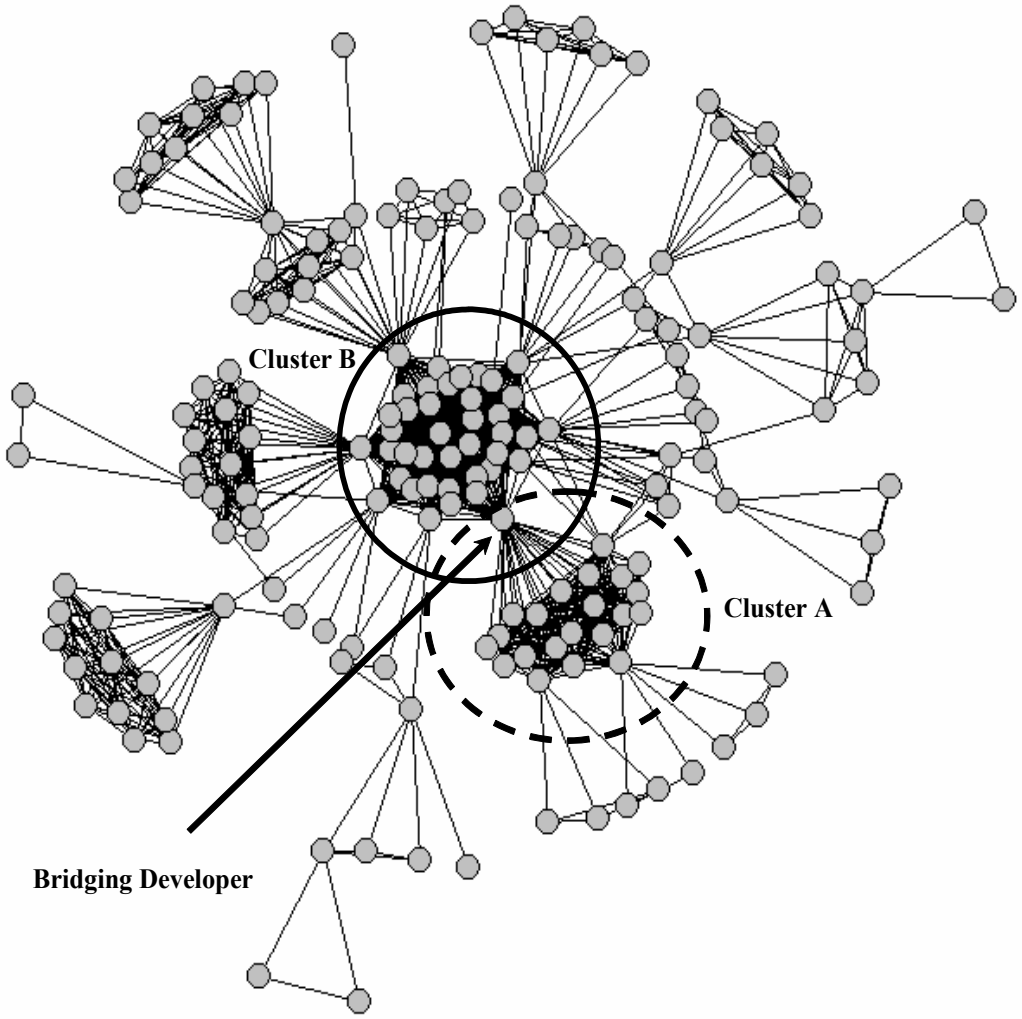
- Travers & Milgram 1969: classic early social network study
  - Destination: a Boston stockbroker; lived in Sharon, MA
  - Sources: Nebraska stockowners; Nebraska and Boston “randoms”
  - Forward letter to a first-name acquaintance “closer” to target
  - Target information provided: name, address, occupation, firm, college, wife’s name and hometown

- Basic findings:
  - 64 of 296 chains reached the target
  - Average length of completed chains: 5.2

FOSTER SCHOOL OF BUSINESS

How Does a Small World look?

- Small World Structure:
- Several clusters of individuals loosely connected with each other.



FOSTER SCHOOL OF BUSINESS

The Strength of Weak Ties

- Not all links are of equal importance

- Granovetter 1974: study of job searches
  - 56% found current job via a personal connection
  - Of these, 16.7% saw their contact “often”
  - The rest saw their contact “occasionally” or “rarely”

- Your “closest” contacts might not always be the most useful: Why?
  - Similar backgrounds and experience
  - They may not know much more than you do

FOSTER SCHOOL OF BUSINESS

From Small Worlds & Weak Ties to Recommenders & Virality

- Many real social networks are "small worlds":
  - High clustering (tight local groups) + short paths between any two people
- Weak ties (acquaintances) are powerful for spreading new information:
  - Job opportunities, product ideas, viral content often travel through weak ties
- In digital platforms, these ideas show up in:
  - Recommender systems
    - "People like you in other clusters also watched/bought…"
    - Bridges between communities help surface diverse content & products.
  - Viral marketing & influence campaigns
  - Seeding not only high-degree "hubs" but also brokers who connect different groups.
  - AI models on graphs
    - Use neighborhoods and paths in the network to predict who will adopt, share, or churn.

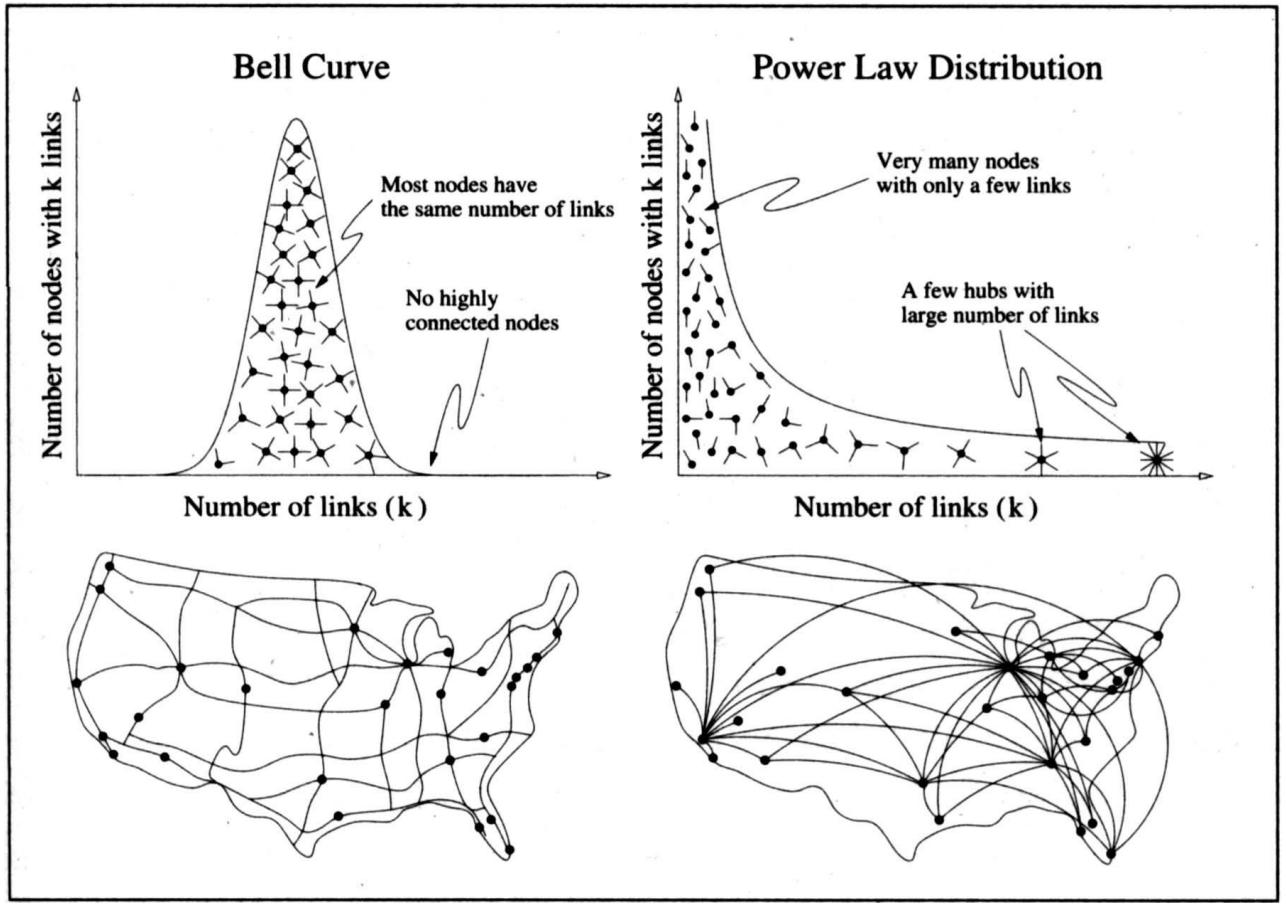
FOSTER SCHOOL OF BUSINESS

Scale Free Networks

- A small proportion of the nodes in a scale-free network have a high degree of connection
- Power law distribution
  - A given node has k connections to other nodes with probability as the power law distribution with exponent $\gamma \sim [2, 3]$ (k to the power of - γ)
  - Hubs and long tail
- Examples of known scale-free networks:
  - Communication networks - Internet
  - Ecosystems and cellular systems
  - Social networks responsible for spread of disease
  - Friendship networks
  - Trust networks

FOSTER SCHOOL OF BUSINESS

Scale Free Networks



Reprinted from Linked: The New Science of Networks by Albert-Laszlo Barabasi

FOSTER SCHOOL OF BUSINESS

Scale Free Networks: 80-20 Rule

- Given by Pareto
  - 80 percent of his peas were produced by 20 percent of his peapods
  - 80 percent of land in Italy was owned by 20 percent population
  - 80% of profit comes from 20% of employees
  - 80% of decisions are made in 20% of meeting time

- On the web
  - 80% of the web links point to 20% of web pages
  - 80% of citations point to 20% of scientists
  - 80% of links in Hollywood are connected to 20% of actors

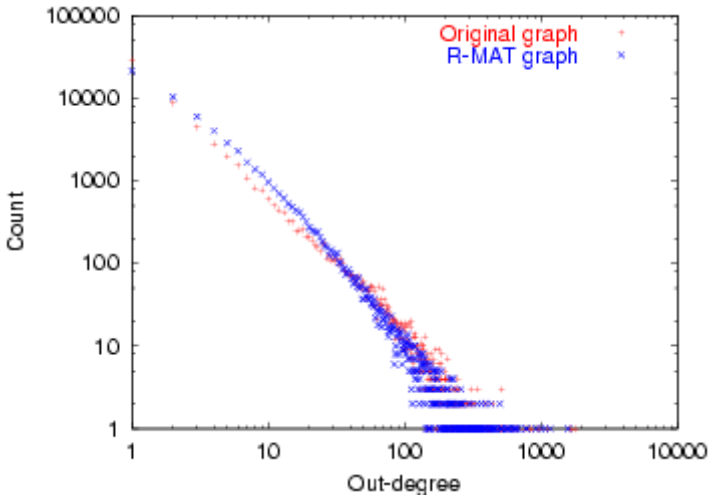
The 80-20 rule is not a universal truth and may not perfectly apply to every single situation, and sometimes the distribution of effort and results can be more evenly distributed.

FOSTER SCHOOL OF BUSINESS

epinions.com

- who-trusts-whom [Richardson + Domingos, KDD 2001]

count



(out) degree

trusts-2000-people user

Hubs, Robustness & Misinformation: Why Scale-Free Networks Matter

- Many online networks are scale-free: few hubs have huge degree, most few links.
- Implications for platform design and AI:
  - Robustness & risk
    - Random failures (removing random users/servers) often have little effect.
    - Targeted attacks on hubs (key accounts, central servers) can fragment the network.
  - Epidemics & information spread
    - Hubs accelerate spread of viruses, memes, and misinformation.
    - Models of epidemics on networks (diffusion models) often assume a scale-free structure.
- In practice, platforms and AI systems must decide:
  - How much visibility and power to give super-hubs (influencers, big pages).
  - How to detect and moderate harmful hubs (botnets, misinformation superspreaders).
  - How to design algorithms that don’t make the rich richer by over-rewarding popularity.

FOSTER SCHOOL OF BUSINESS

Cohesive Subgroups

- Cohesive subgroups are subsets of actors among whom there are relatively strong, direct, intense, frequent, or positive ties.
- Although there are numerous ways to conceptualize the idea of subgroup, there are four general properties:
  - The mutuality of ties
  - The closeness of reachability of subgroup members
  - The frequency of ties among members
  - The relative frequency of ties among subgroup members compared to non-members

FOSTER SCHOOL OF BUSINESS

Subgroups Based on Complete Mutuality

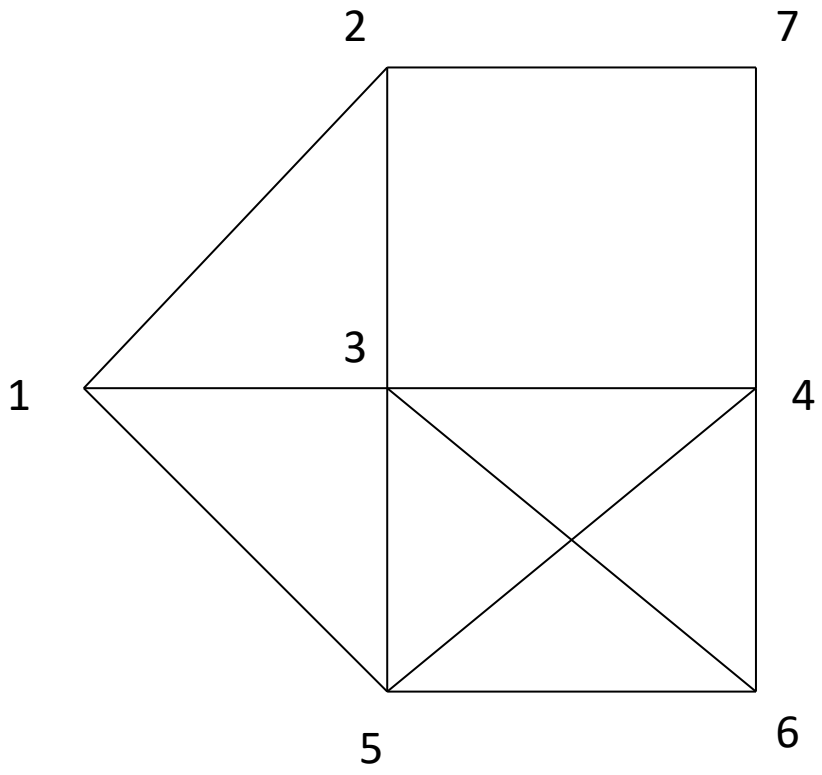
- **"Cliquish"** subgroups
  - Cohesive subgroups in relations would be characterized by sets of people among whom all friendship choices were mutual.

- Definition of a Clique
  - A clique in a graph is a **maximal complete** subgraph of three or more nodes, all of which are adjacent to each other, and there are no other nodes that are also adjacent to all of the members of the clique.

FOSTER SCHOOL OF BUSINESS

Subgroups Based on Complete Mutuality

- Example



Cliques: {1,2,3},{1,3,5}, and {3,4,5,6}

Communities in Practice: From Cliques to Community Detection & Graph ML

- Networks have cohesive groups:
  - Cliques and dense clusters → tight communities, interest groups, social circles.
  - Core-periphery structure → highly connected "core" + loosely connected "periphery."
- In practice, we don't just eyeball cliques, we use algorithms:
  - Community detection (e.g., Louvain, label propagation) to find groups at scale
  - Graph embeddings to cluster nodes into communities in vector space
- These communities + SNA metrics become inputs to AI systems:
  - Marketing & product:
    - Define segments for targeting, personalization, and A/B testing.
    - Identify niches vs mainstream core for positioning & messaging.
  - Platform health:
    - Detect suspicious clusters (fraud rings, bot farms)
    - Monitor how content spreads across communities (echo chambers vs cross-talk)
- Centrality + communities are often features in graph ML models:
  - Node features: degree, betweenness, community ID, core/periphery role.
  - Help models learn better representations and improve interpretability.

FOSTER SCHOOL OF BUSINESS

From SNA to Graph Machine Learning

- Social networks are graphs:
  - nodes (people, pages, items) and edges (relationships, interactions).
- Classic SNA asks:
  - who is important, how are groups structured, how does information flow?
- Graph machine learning uses these graphs as inputs to predictive models:
  - Node classification (e.g., churn risk, bot detection, lead scoring)
  - Link prediction (e.g., friend and follower suggestions, "people you may know")
  - Recommendation on user-item graphs (e.g., "users who liked X also liked Y")
  - Graph classification (e.g., detecting fraud rings or coordinated campaigns)

SNA gives us intuition and features. Graph ML scales that intuition with modern ML and AI models.

FOSTER SCHOOL OF BUSINESS

Graph Embeddings: Making Nodes ML-Friendly

- Goal: represent each node as a dense vector (embedding) that captures its position in the network.
- Idea (DeepWalk / node2vec):
  - Simulate random walks in the graph (like "sentences" of nodes).
  - Treat nodes like words and learn embeddings with word2vec-style models.
  - Nodes that appear in similar walk contexts get similar vectors.
- Result: nodes with similar roles, neighbors, or communities are close in embedding space.
- Uses in practice:
  - Cluster users or pages into segments.
  - Feed node embeddings into downstream models for prediction or recommendation.
  - Visualize the graph in 2D by reducing embedding dimensions (e.g., t-SNE, UMAP).

FOSTER SCHOOL OF BUSINESS

Graph Neural Networks (GNNs) at a Glance

- GNNs generalize neural networks to graph-structured data.
- Key idea: a node's representation is updated by aggregating information from its neighbors.
- Typical GNN layer:
  - Start with node features (e.g., profile, text, simple SNA metrics).
  - Aggregate neighbor features (e.g., mean, weighted sum, attention).
  - Apply a neural network to produce a new node representation.
- Stack multiple layers to capture information from multi-hop neighborhoods.
- Applications in industry:
  - Recommendation (Pinterest, TikTok, YouTube user-item graphs).
  - Fraud and spam detection in social and financial networks.
  - Content moderation and abuse detection based on network patterns.

FOSTER SCHOOL OF BUSINESS

Where SNA Fits into AI on Graphs

- Classic SNA concepts remain useful in the age of graph ML:
  - Centrality (degree, betweenness, eigenvector) becomes input features for ML models.
  - Community structure (clusters, core-periphery) informs segmentation and sampling.
  - Path lengths and connectivity inform diffusion and recommendation strategies.
- Interpretability: SNA offers human-friendly explanations of complex graph models.
  - "Model pays special attention to high-betweenness nodes" → brokers and bridges.
  - "Embeddings cluster around dense communities" → interest or social groups.
- For digital marketing and information systems, combining SNA + graph ML enables:
  - Influencer identification and campaign seeding
  - Network-aware churn and upsell models
  - Richer recommendation and personalization systems

FOSTER SCHOOL OF BUSINESS

Ethics and Bias in Graph Machine Learning

- Real-world networks reflect existing social biases and inequalities.
- Graph ML models can amplify these patterns:
  - Popularity bias: "rich get richer”: already popular nodes receive even more exposure.
  - Homophily: people connect to people like themselves, which can entrench segregation.
  - Minority or fringe groups may be under-recommended or over-flagged.
- Risks in digital marketing and platforms:
  - Over-targeting influential nodes can widen visibility gaps.
  - Content moderation models may unfairly impact certain communities.
- Responsible practices:
  - Monitor for disparate impacts across groups.
  - Incorporate fairness constraints and diverse training data.
  - Use SNA metrics and audits to understand who benefits and who is left out.

FOSTER SCHOOL OF BUSINESS